

## Task 6: Timer Interrupt

6\_timer\_interrupt.ino

```
1  #define timeSeconds 10
2  // Set GPIOs for LED and PIR Motion Sensor
3  const int led = 26;
4  const int motionSensor = 27;
5  // Timer: Auxiliary variables
6  unsigned long now = millis();
7  unsigned long lastTrigger = 0;
8  boolean startTimer = false;
9  boolean motion = false;
10 // Checks if motion was detected, sets LED HIGH and starts a timer
11 void IRAM_ATTR detectsMovement() {
12     digitalWrite(led, HIGH);
13     startTimer = true;
14     lastTrigger = millis();
15 }
16 void setup() {
17     // Serial port for debugging purposes
18     Serial.begin(115200);
19
20     // PIR Motion Sensor mode INPUT_PULLUP
21     pinMode(motionSensor, INPUT_PULLUP);
22     // Set motionSensor pin as interrupt, assign interrupt function and set RISING mode
23     attachInterrupt(digitalPinToInterrupt(motionSensor), detectsMovement, RISING);
24     // Set LED to LOW
25     pinMode(led, OUTPUT);
26     digitalWrite(led, LOW);
27 }
28 void loop() {
29     // Current time
30     now = millis();
31     if((digitalRead(led) == HIGH) && (motion == false)) {
32         Serial.println("MOTION DETECTED!!!");
33         motion = true;
34     }
35     // Turn off the LED after the number of seconds defined in the timeSeconds variable
36     if(startTimer && (now - lastTrigger > (timeSeconds*1000))) {
37         Serial.println("Motion stopped...");
38         digitalWrite(led, LOW);
39         startTimer = false;
40         motion = false;
41     }
42 }
43
```